TEAM NUMBER: 165
POCKET PLAY: GAMIFIED FINANCIAL LITERACY PLATFORM
Final Technical & Impact Report | Vanguard India Challenge (Megathon 2026)
Live Application: https://pocket-play.onrender.com/

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SECTION 1: SOLUTION OVERVIEW
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Target Problem:
Conventional financial education for Grade 11–12 students in underserved areas of Hyderabad relies almost exclusively on abstract, lecture-based theory. This results in high student drop-off rates and fails to develop practical, intuitive financial habits before students enter legal adulthood. The Vanguard India Challenge requires an engaging, non-commercial digital platform designed strictly to educate without promoting speculative stock tips, tailored for low-resource public school computer labs.

Our Solution — "The Parallel Realm Progression":
Pocket Play reframes financial literacy through an interactive world-clearing journey. The player awakens in a strange parallel realm stripped of modern financial systems and discovers that returning home requires mastering the economic rules governing this reality. Starting at Rupee Zero (₹0), the player must unlock and clear distinct thematic worlds by conquering practical economic trials. Rather than answering passive multiple-choice quizzes, students learn by directly executing financial decisions, confronting immediate graphical balance adjustments, and iterating through risk-free educational retry loops.

Coverage of Vanguard's Four Learning Pillars:
1. Plan & Budget: 50/30/20 framework allocation, unit price calculations, and student cash-flow balancing.
2. Save & Manage: 3–6 month emergency reserve buffers, banking instruments (Savings vs. RD/FD), and absorbing expense shocks.
3. Understand Investing: Stock equity ownership, P/E valuation, broad-market Index ETFs vs. single-stock risk, and Rule of 72 compounding.
4. Stay Financially Safe: Digital defense against predatory informal credit (60% APR traps), OTP phishing, and UPI request-money scams.

Accessibility & Lab Viability:
- Zero-PII Anonymous Sessions: Eliminates login friction via 6-character recovery codes (e.g., HYD-743).
- Instant Trilingual Localization: Native runtime translation across English, Telugu, and Hindi without page reloads.
- NGO Cohort Telemetry: Real-time cohort monitoring for workshop facilitators without student tracking.

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SECTION 2: APPROACH & ARCHITECTURE
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1. Architecture Stack:
- Client Tier: React, TypeScript, Vite, Tailwind CSS. Mathematical simulations (stock volatility cycles, compounding curves, loan amortizations) execute client-side, ensuring zero latency on shared school lab bandwidth.
- Application Tier: Lightweight Node.js (v20 LTS) Express service exposing high-throughput REST APIs (/api/session/*, /api/telemetry/*, /api/facilitator/*).
- Persistence Layer: Embedded SQLite (game.db) compiled natively from source on Linux, using atomic COALESCE update queries to prevent concurrency race conditions.

- Cloud Hosting: Fully containerized and deployed on Render (https://pocket-play.onrender.com/).

2. Zero-PII Session Lifecycle:
To protect student privacy across public school labs, students start instantly without email or phone verification. The server generates a unique deterministic token (e.g., HYD-743), persisting wallet balance, emergency reserve, initial CIBIL score (650), language choice, and unlocked realms in SQLite. Students can resume sessions on any lab computer seamlessly.

3. Four-Stage Educational Game Loop:
- Step 1 (Context Dilemma): Everyday student dilemma (e.g., speculative day-trading hype vs. long-term disciplined investing).
- Step 2 (Micro-Lesson): Concise 30-second breakdown of the core financial mechanism.
- Step 3 (Simulation Sandbox): Interactive slider/sorting interface with dynamic visual feedback.
- Step 4 (Consequence & Evaluation): Immediate balance updates (₹ cash, portfolio value, CIBIL score) paired with zero-penalty retry paths.

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SECTION 3: CORE FEATURES & USER MODES
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1. Dual Progression Modes:
- Story Mode: Sequential narrative progression starting with a 3-minute survival trial covering barter, the origin of currency (₹), and cash reserves, unlocking successive realms.
- Custom Mode (Skill Tree / Flowchart): An unblocked node map allowing teachers and NGO workshop leads to launch directly into specific simulation sandboxes for focused 30-minute modules.

2. Trilingual Localization Layer:
Full native support for English, Telugu (తెలుగు), and Hindi (हिन्दी). Text swaps occur instantaneously without full-page reloads, persisting language preference directly to the backend session state.

3. Interactive Simulation Portals:
- Frontier Outpost (Plan & Budget): 50/30/20 budget slider balancing daily stipends against essential needs (bus pass, books) vs. discretionary wants.
- Citadel of Debt (Save & Manage): Compares informal moneylenders (5%/month = 60% APR) against regulated credit, demonstrating debt traps and dynamic CIBIL score shifts.
- Grand Exchange (Stocks & Investing): Interactive stock market simulator teaching partial business ownership, P/E ratio valuation (identifying overhyped stocks), dividends, and SEBI regulatory protections. Demonstrates why diversified broad-market Index ETFs (Nifty 50) protect against market volatility and outperform single-stock gambles over 5-to-20 year horizons.
- Cyber Frontier (Stay Financially Safe): Phishing gauntlet simulating SMS and UPI requests, reinforcing the golden rule: Entering your UPI PIN always deducts money.

4. Facilitator Telemetry Dashboard:
Exposes an aggregated endpoint (/api/facilitator/cohort-summary) tracking cohort participants, average savings reserves, median CIBIL scores, and common fraud errors without collecting student PII.

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SECTION 4: RESULTS, TESTING & DEPLOYMENT
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1. Reliability & Testing:

- Integer Arithmetic: Strictly enforced integer math for Indian Rupees (₹) to eliminate floating-point rounding drift across multi-year compounding calculations.
- Concurrency Testing: Verified serialized SQLite transactions under simulated multi-user concurrent read/write loads.
- Lightweight Footprint: Optimized bundle size loading in <800ms on budget Android devices and throttled 3G networks.
- Educational Safety: Tested fail-safe feedback loops where erroneous financial decisions (such as panic-selling during a stock market correction) trigger constructive explanations rather than punishing dead ends.

2. Alignment with Vanguard Measures of Success:
- Simple: Replaced complex Wall Street jargon with visual sliders and everyday relatable metaphors.
- Relatable: Grounded in Indian Rupees (₹) with scenarios youth face (canteen splits, phone EMIs, UPI QR codes).
- Language-Ready: Zero-latency runtime switching across English, Telugu, and Hindi.
- Persona-Led: Calibrated for Grade 11–12 students entering financial independence.
- Game-Based: Practical scenario challenges replacing passive quizzes.
- NGO-Integrated: 6-character lab recovery tokens and live facilitator cohort analytics.

3. Live Deployment:
The application is live 24/7 on Render:
https://pocket-play.onrender.com/